WORKED SOLUTION

STEP III 1999, Question 5

Pure • Fibonacci Identities and Induction

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/99-s3-q5

The key idea

This is the Fibonacci sequence, shifted so that $u_n = F_{n+1}$. For the first identity, write both u_{n+2} and u_{n-1} in terms of the middle pair u_{n+1} and u_n: the cross terms then cancel. For the doubling formulae, prove the two statements *together* by induction, because each one feeds the other through the recurrence. The first identity is exactly the tool needed to finish the odd case.

Solution

The first identity

Fix $n \geq 1$, so that u_{n-1}, u_n, u_{n+1}, u_{n+2} are all defined. The recurrence gives

$$u_{n+2} = u_{n+1} + u_n \qquad \text{and} \qquad u_{n-1} = u_{n+1} - u_n,$$

the second coming from $u_{n+1} = u_n + u_{n-1}$. Squaring and adding,

$$\begin{aligned} u_{n+2}^2 + u_{n-1}^2 &= (u_{n+1}^2 + 2u_{n+1}u_n + u_n^2) + (u_{n+1}^2 - 2u_{n+1}u_n + u_n^2) \\ &= 2(u_{n+1}^2 + u_n^2), \end{aligned}$$

which is the required identity.

The doubling formulae

Write $P(n)$ for the statement that *both*

$$u_{2n} = u_n^2 + u_{n-1}^2 \qquad \text{and} \qquad u_{2n+1} = u_{n+1}^2 - u_{n-1}^2.$$

The first terms are $u_0, u_1, \cdots = 1, 1, 2, 3, 5, 8, \ldots$.

Base case. For $n = 1$: $u_2 = 2 = 1^2 + 1^2 = u_1^2 + u_0^2$, and $u_3 = 3 = 2^2 - 1^2 = u_2^2 - u_0^2$. So $P(1)$ holds.

Ask yourself Why prove the two formulae together rather than one at a time?

To get u_{2n+2} from the recurrence we need u_{2n+1} and u_{2n}, so the even formula at $n+1$ needs *both* formulae at n. The odd formula at $n+1$ then needs the even one at $n+1$ as well. Neither statement can be pushed forward on its own.

Inductive step. Suppose $P(n)$ holds for some $n \geq 1$.

Even term. By the recurrence and then $P(n)$,

$$\begin{aligned} u_{2n+2} &= u_{2n+1} + u_{2n} \\ &= (u_{n+1}^2 - u_{n-1}^2) + (u_n^2 + u_{n-1}^2) \\ &= u_{n+1}^2 + u_n^2, \end{aligned}$$

which is the even formula with n replaced by $n+1$.

Odd term. Using the recurrence, the line just proved and $P(n)$,

$$\begin{aligned} u_{2n+3} &= u_{2n+2} + u_{2n+1} \\ &= (u_{n+1}^2 + u_n^2) + (u_{n+1}^2 - u_{n-1}^2) \\ &= 2u_{n+1}^2 + u_n^2 - u_{n-1}^2. \end{aligned}$$

We want this to equal $u_{n+2}^2 - u_n^2$, and here the first identity comes in. Since $n \geq 1$, it gives $u_{n+2}^2 = 2u_{n+1}^2 + 2u_n^2 - u_{n-1}^2$, so

$$u_{n+2}^2 - u_n^2 = 2u_{n+1}^2 + u_n^2 - u_{n-1}^2 = u_{2n+3}.$$

This is the odd formula with n replaced by $n+1$, since $2(n+1)+1 = 2n+3$ and $(n+1) \pm 1 = n+2,\ n$.

So $P(n)$ implies $P(n+1)$. With the base case, $P(n)$ holds for every positive integer n by induction.

Checks

Try $n = 5$: $u_{10} = 89$ and $u_5^2 + u_4^2 = 64 + 25 = 89$; $u_{11} = 144$ and $u_6^2 - u_4^2 = 169 - 25 = 144$. We checked the first identity for all $n \leq 299$, and both doubling formulae for all $n \leq 189$, in exact integer arithmetic. Note also that $u_{n+1}^2 - u_{n-1}^2 = (u_{n+1} - u_{n-1})(u_{n+1} + u_{n-1}) = u_n(u_{n+1} + u_{n-1})$, a handy alternative form for the odd terms.

Common slips

- Inducting on the even formula alone: the step needs u_{2n+1}, so it stalls.
- Using the first identity at $n = 0$, where u_{-1} is not defined. In the inductive step we only use it for $n \geq 1$.
- Forgetting that the indexing is shifted from the usual Fibonacci numbers ($u_0 = u_1 = 1$), and checking the base case with the wrong values.

Beyond STEP

The doubling formulae let you compute u_N in about $\log_2 N$ steps, by halving N repeatedly: this is the "fast doubling" method, a cousin of fast exponentiation. Behind it sits the matrix identity $\begin{pmatrix} 1 & 1 \\ 1 & 0 \end{pmatrix}^n = \begin{pmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{pmatrix}$, and squaring this matrix gives both formulae at once. Strang's *Introduction to Linear Algebra* treats Fibonacci numbers through powers and eigenvalues of this matrix, and the chapter on continued fractions in Davenport's *The Higher Arithmetic* is where ratios of consecutive Fibonacci numbers reappear, as the convergents of the golden ratio.

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